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Course for the Faculties of Sciences and Chemical Sciences

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Universidad de Salamanca

1st Edition • 2026

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Welcome to the **Solid State Physics Notes**.

What is this?

This electronic book compiles the main topics and problems from the Solid State Physics course at the University of Salamanca (USAL). Here you will find detailed explanations and solved problems designed to facilitate learning and understanding of key concepts in the electronic structure of matter.

Available Content

This interactive edition includes the following topics and problems:

- **1.1 Introduction:** Fundamental concepts of solid state physics.
- **1.2 Calculation of Band Structure:** Models and methodologies to determine energy bands in solids.
- **1.3 Tight Binding Method:** Theory and LCAO approximation.
- **1.3 Problem: sp³ Model:** Application of the tight binding method in sp³ hybridizations.
- **1.3 Problem: Graphene:** Study of the graphene band structure using Tight Binding.
- **1.4 Pseudopotentials:** Fundamentals and usefulness of the pseudopotential approximation.
- **1.4 Problem: Pseudopotential:** Practical application of the method in crystal lattices.

PDF Version

You can also download the printable, formatted version of the book:

- [Download PDF in English](#)
- [Descargar PDF en español](#)

Note

This interactive project is optimized for web viewing and PDF compilation, preserving the highest typographical quality using LaTeX.

Part I

Solid State Physics Notes

1.1 INTRODUCTION

(Translation pending)

1.2 CALCULATION OF BAND STRUCTURE

(Translation pending)

1.3 TIGHT BINDING

(Translation pending)

1.3 PROBLEM: SP3 MODEL

(Translation pending)

1.3 PROBLEM: GRAPHENE

(Translation pending)

1.4 PSEUDOPOTENTIAL

(Translation pending)

1.4 PROBLEM: PSEUDOPOTENTIAL

(Translation pending)

Part II

References

BIBLIOGRAPHY